

Medical AI Suite — Persisted model evaluation

Persisted model evaluation report

Evaluation used saved production pipelines on recreated stratified 80/20 holdout test sets (‘random_state=42’). No

Liver model evaluation

Selected model: random_forest
Train/test: 466/117

Metric	Value
Accuracy	0.7436
Balanced Accuracy	0.6630
Precision	0.7978
Recall	0.8554
F1	0.8256
Roc Auc	0.7697
Specificity	0.4706
Sensitivity	0.8554
Mcc	0.3469
Cohen Kappa	0.3439
Log Loss	0.5031
Brier Score	0.1721

Confusion matrix: TN=16, FP=18, FN=12, TP=71

precision	recall	f1-score	support	0	0.57	0.47	0.52	34
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Threshold	Precision	Recall	F1
0.3	0.7248	0.9518	0.8229
0.4	0.7475	0.8916	0.8132
0.5	0.7912	0.8675	0.8276
0.6	0.8182	0.7590	0.7875
0.7	0.9216	0.5663	0.7015

Diabetes model evaluation

Selected model: svm
Train/test: 614/154

Metric	Value
Accuracy	0.7273
Balanced Accuracy	0.7261
Precision	0.5909
Recall	0.7222
F1	0.6500
Roc Auc	0.8139
Specificity	0.7300
Sensitivity	0.7222
Mcc	0.4360
Cohen Kappa	0.4302
Log Loss	0.4994
Brier Score	0.1673

Confusion matrix: TN=73, FP=27, FN=15, TP=39

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	precision	recall	f1-score	support
...	0.83	0.73	0.78	100

Threshold	Precision	Recall	F1
0.3	0.5844	0.8333	0.6870
0.4	0.6230	0.7037	0.6609
0.5	0.6596	0.5741	0.6139
0.6	0.6571	0.4259	0.5169
0.7	0.7273	0.2963	0.4211

Limitations
Public datasets are relatively small and metrics are not clinical evidence. Results may not generalize across popul